

MEDICAL INVENTORY MANAGEMENT

Date	20-06-2025
Team ID	LTVIP2025TMID29505
Project Name	Medical Inventory Management
Maximum Marks	

CHAPTER - 4 PROJECT DESIGN

4.1 - EXISTING SYSTEM (PROBLES SOLUTION FIT):

Here's a clear explanation of the Existing System and Problem-Solution Fit for your Medical Inventory Management System using Salesforce project:

Existing System Overview

Most traditional medical inventory systems are either manual or built on basic desktop applications. These systems typically involve:

- Paper-based logs or spreadsheets for tracking stock
- Time-consuming manual updates and audits
- High risk of human error in expiry tracking and batch identification
- Limited data security and access control
- Poor visibility into real-time inventory across multiple locations

Common Challenges:

Problem	Impact
Manual tracking	Delays in replenishment and increased errors
No expiry alerts	Risk of using expired medicines
Fragmented systems	Lack of centralized data and reporting
No automation	Increased workload for staff
Poor compliance	Difficulty during audits and regulatory checks

Problem-Solution Fit with Salesforce

Salesforce offers a cloud-based, scalable, and customizable platform that directly addresses these pain points:

Challenge	Salesforce Solution
Manual tracking	Real-time inventory dashboards and mobile access
Expiry management	Automated alerts and batch tracking
Fragmented systems	Unified platform with integration via MuleSoft
Lack of automation	Salesforce Flow for reordering and notifications
Compliance issues	Audit trails, role-based access, and secure data handling

4.2 - PROPOSED SYSTEM (PROPOSED SOLUTION):

The proposed system is a cloud-based Medical Inventory Management Solution built on the Salesforce platform, designed to automate and optimize inventory operations across healthcare facilities. It replaces manual tracking and fragmented systems with a centralized, intelligent, and scalable solution that ensures real-time visibility, regulatory compliance, and operational efficiency.

Key Features of the Proposed Solution

Feature	Description
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MEDICAL INVENTORY MANAGEMENT

Real-Time Inventory Tracking	Monitor stock levels, batch numbers, and expiry dates across multiple locations
Automated Reordering	Trigger purchase orders when stock falls below threshold using Salesforce Flow
Supplier Management	Maintain vendor profiles, order history, and delivery performance
Expiry & Compliance Alerts	Notify staff of upcoming expirations and ensure audit readiness
Mobile Access	Allow field reps and pharmacists to update inventory from hospital sites
Reports & Dashboards	Visualize inventory trends, usage analytics, and procurement efficiency

4.3-SOLUTION ARCHITECTURE

- a. **Scalability:** The architecture should be designed to accommodate future growth and changes in inventory needs.
- b. **Flexibility:** It should be flexible enough to adapt to different healthcare settings and workflows.
- c. **Integration:** Seamless integration with existing healthcare IT systems is crucial.